The Potential Outcome Framework, Part 1

Stat 256, Causality

Chad Hazlett

UCLA

chazlett@ucla.edu

Spring 2026

Readings

Chapter 1-3 from Hernan & Robins, Causal Inference: What If.

Outline

1 Potential Outcomes

The Neyman-Rubin Potential Outcome Model

A considerable part of the empirically oriented causal inference community works mainly in terms of the Neyman-Rubin causal model, aka the Potential Outcomes Model (POM).

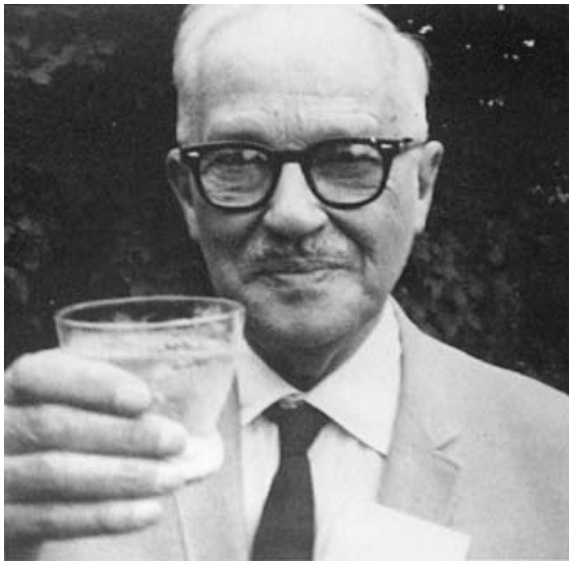


Figure: Neyman

Figure: Rubin

The Potential Outcomes Approach in a Nutshell

What do we mean when we say something had a causal effect?

The Potential Outcomes Approach in a Nutshell

What do we mean when we say something had a causal effect?

The POM answer:

- Suppose unit i takes a treatment and you measure the outcome. Let's call this Y_{1i}, or the treatment potential outcome

The Potential Outcomes Approach in a Nutshell

What do we mean when we say something had a causal effect?

The POM answer:

- Suppose unit i takes a treatment and you measure the outcome. Let's call this Y_{1i}, or the treatment potential outcome
- You'd like to know how Y_{1i} compares to the value of Y unit i *would have had, if it had not taken the treatment*. Call this Y_{0i}, or the non-treatment potential outcome

The Potential Outcomes Approach in a Nutshell

What do we mean when we say something had a causal effect?

The POM answer:

- Suppose unit i takes a treatment and you measure the outcome. Let's call this Y_{1i}, or the treatment potential outcome
- You'd like to know how Y_{1i} compares to the value of Y unit i *would have had, if it had not taken the treatment*. Call this Y_{0i}, or the non-treatment potential outcome
- Then unit i's *treatment effect* is $\tau_i = Y_{1i} - Y_{0i}$.

The Potential Outcomes Approach in a Nutshell

What do we mean when we say something had a causal effect?

The POM answer:

- Suppose unit i takes a treatment and you measure the outcome. Let's call this Y_{1i}, or the treatment potential outcome
- You'd like to know how Y_{1i} compares to the value of Y unit i *would have had, if it had not taken the treatment*. Call this Y_{0i}, or the non-treatment potential outcome
- Then unit i's *treatment effect* is $\tau_i = Y_{1i} - Y_{0i}$.
- You do not get to see both Y_{1i} and Y_{0i} for any unit i. This is the fundamental problem of causal inference.

The Potential Outcomes Approach in a Nutshell

What do we mean when we say something had a causal effect?

The POM answer:

- Suppose unit i takes a treatment and you measure the outcome. Let's call this Y_{1i}, or the treatment potential outcome
- You'd like to know how Y_{1i} compares to the value of Y unit i *would have had, if it had not taken the treatment*. Call this Y_{0i}, or the non-treatment potential outcome
- Then unit i's *treatment effect* is $\tau_i = Y_{1i} - Y_{0i}$.
- You do not get to see both Y_{1i} and Y_{0i} for any unit i. This is the fundamental problem of causal inference.
- Learning anything about causality requires somehow filling in the "missing" potential outcomes through some set of assumptions.

Fundamental Quantities (so far)

- Y_{1i} and Y_{0i}, the potential outcomes (also written $Y_i(1)$, $Y_i(0)$, $Y_i(d)$).
- D_i, the treatment status of unit i
- Observed outcome, $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$
- $\tau_i = Y_{1i} - Y_{0i}$

Fundamental Quantities (so far)

- Y_{1i} and Y_{0i}, the potential outcomes (also written $Y_i(1)$, $Y_i(0)$, $Y_i(d)$).
- D_i, the treatment status of unit i
- Observed outcome, $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$
- $\tau_i = Y_{1i} - Y_{0i}$

Which of the above are observable?

Fundamental Quantities (so far)

- Y_{1i} and Y_{0i}, the potential outcomes (also written $Y_i(1)$, $Y_i(0)$, $Y_i(d)$).
- D_i, the treatment status of unit i
- Observed outcome, $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$
- $\tau_i = Y_{1i} - Y_{0i}$

Which of the above are observable?

Name games: “counterfactuals” vs. “potential outcomes”

POs and QOIs

Some quantities of interest will be:

POs and QOIs

Some quantities of interest will be:

- Average treatment effect (ATE): $\mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[\tau]$

POs and QOIs

Some quantities of interest will be:

- Average treatment effect (ATE): $\mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[\tau]$
- Average treatment effect among the treated (ATET)

$$\mathbb{E}[\tau_i | D_i = 1] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1]$$

POs and QOIs

Some quantities of interest will be:

- Average treatment effect (ATE): $\mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[\tau]$
- Average treatment effect among the treated (ATET)

$$\mathbb{E}[\tau_i | D_i = 1] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1]$$

- Average treatment effect among the controls (ATEC)

$$\mathbb{E}[\tau_i | D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 0]$$

POs and QOIs

Some quantities of interest will be:

- Average treatment effect (ATE): $\mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[\tau]$
- Average treatment effect among the treated (ATET)

$$\mathbb{E}[\tau_i | D_i = 1] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1]$$

- Average treatment effect among the controls (ATEC)

$$\mathbb{E}[\tau_i | D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 0]$$

- Conditional average treatment effect (CATE):

$$\mathbb{E}[\tau_i | X = x] = \mathbb{E}[Y_{1i} - Y_{0i} | X = x]$$

Embodied assumptions

Just by writing down POs using notation $Y_i(d)$ we assert:

- consistency: $Y_i(d)$ is the Y_i you'd see if i received treatment d.
- no-interference: we've only asked if unit i took the treatment, not unit j, so presumably $Y_i(D_i, D_j) = Y_i(D_i)$ regardless of the value of D_j.
- one-version-of-treatment: the $D = d$ means the same thing all the time (though this is sketchy)

Embodied assumptions

Just by writing down POs using notation $Y_i(d)$ we assert:

- consistency: $Y_i(d)$ is the Y_i you'd see if i received treatment d.
- no-interference: we've only asked if unit i took the treatment, not unit j, so presumably $Y_i(D_i, D_j) = Y_i(D_i)$ regardless of the value of D_j.
- one-version-of-treatment: the $D = d$ means the same thing all the time (though this is sketchy)

Embodied assumptions

Just by writing down POs using notation $Y_i(d)$ we assert:

- consistency: $Y_i(d)$ is the Y_i you'd see if i received treatment d.
- no-interference: we've only asked if unit i took the treatment, not unit j, so presumably $Y_i(D_i, D_j) = Y_i(D_i)$ regardless of the value of D_j.
- one-version-of-treatment: the $D = d$ means the same thing all the time (though this is sketchy)

Consistency is really definitional of our POs. But we name it so we can:

- reference it later (e.g. to say $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_i|D_i = 1]$, and
- remind ourselves it can breakdown, e.g. due to interference.

Embodied assumptions

Just by writing down POs using notation $Y_i(d)$ we assert:

- consistency: $Y_i(d)$ is the Y_i you'd see if i received treatment d.
- no-interference: we've only asked if unit i took the treatment, not unit j, so presumably $Y_i(D_i, D_j) = Y_i(D_i)$ regardless of the value of D_j.
- one-version-of-treatment: the $D = d$ means the same thing all the time (though this is sketchy)

Consistency is really definitional of our POs. But we name it so we can:

- reference it later (e.g. to say $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_i|D_i = 1]$, and
- remind ourselves it can breakdown, e.g. due to interference.

Non-interference + one-version are usually stated as part of Stable Unit Treatment Value assumption (SUTVA, Rubin 1980)

Embodied assumptions

Just by writing down POs using notation $Y_i(d)$ we assert:

- consistency: $Y_i(d)$ is the Y_i you'd see if i received treatment d.
- no-interference: we've only asked if unit i took the treatment, not unit j, so presumably $Y_i(D_i, D_j) = Y_i(D_i)$ regardless of the value of D_j.
- one-version-of-treatment: the $D = d$ means the same thing all the time (though this is sketchy)

Consistency is really definitional of our POs. But we name it so we can:

- reference it later (e.g. to say $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_i|D_i = 1]$, and
- remind ourselves it can breakdown, e.g. due to interference.

Non-interference + one-version are usually stated as part of Stable Unit Treatment Value assumption (SUTVA, Rubin 1980)

We can relax these in various ways, including:

- "treatment variation irrelevance" (VanderWeele 2009)
- expanding POs to index treatment of others in some limited way
- broadening treatment definition

POs can be seductive

Not all POs that can be written are necessarily what you want.

POs can be seductive

Not all POs that can be written are necessarily what you want.

Changing the treatment status of unit *i* to a counterfactual value could affect other factors that could affect our outcome, or may cause our unit to leave the population. That may be want you want, but

- note that it is not about “holding all else constant”
- it sometimes engages absurdities

POs can be seductive

Not all POs that can be written are necessarily what you want.

Changing the treatment status of unit *i* to a counterfactual value could affect other factors that could affect our outcome, or may cause our unit to leave the population. That may be want you want, but

- note that it is not about "holding all else constant"
- it sometimes engages absurdities

Example: what is the effect of leader gender on some outcome?

POs can be seductive

Not all POs that can be written are necessarily what you want.

Changing the treatment status of unit i to a counterfactual value could affect other factors that could affect our outcome, or may cause our unit to leave the population. That may be want you want, but

- note that it is not about "holding all else constant"
- it sometimes engages absurdities

Example: what is the effect of leader gender on some outcome?

- For US presidents, this involves $Y_{trump}(female)$, i.e. "What would Donald Trump have done if he was a woman?'

POs can be seductive

Not all POs that can be written are necessarily what you want.

Changing the treatment status of unit i to a counterfactual value could affect other factors that could affect our outcome, or may cause our unit to leave the population. That may be want you want, but

- note that it is not about "holding all else constant"
- it sometimes engages absurdities

Example: what is the effect of leader gender on some outcome?

- For US presidents, this involves $Y_{trump}(female)$, i.e. "What would Donald Trump have done if he was a woman?'
- That is hard to imagine, depends on when you change his sex and how, affects many other characteristics, affects whether he would have been in sample of presidents, ...

POs can be seductive

Not all POs that can be written are necessarily what you want.

Changing the treatment status of unit i to a counterfactual value could affect other factors that could affect our outcome, or may cause our unit to leave the population. That may be want you want, but

- note that it is not about "holding all else constant"
- it sometimes engages absurdities

Example: what is the effect of leader gender on some outcome?

- For US presidents, this involves $Y_{trump}(female)$, i.e. "What would Donald Trump have done if he was a woman?'
- That is hard to imagine, depends on when you change his sex and how, affects many other characteristics, affects whether he would have been in sample of presidents, ...
- Raises questions as to what causal effect is of real interest.

POs can be seductive

Not all POs that can be written are necessarily what you want.

Changing the treatment status of unit i to a counterfactual value could affect other factors that could affect our outcome, or may cause our unit to leave the population. That may be want you want, but

- note that it is not about "holding all else constant"
- it sometimes engages absurdities

Example: what is the effect of leader gender on some outcome?

- For US presidents, this involves $Y_{trump}(female)$, i.e. "What would Donald Trump have done if he was a woman?'
- That is hard to imagine, depends on when you change his sex and how, affects many other characteristics, affects whether he would have been in sample of presidents, ...
- Raises questions as to what causal effect is of real interest.
- One guide for avoiding this is to assert that a cause must be "manipulable"

POs can be seductive

Not all POs that can be written are necessarily what you want.

Changing the treatment status of unit i to a counterfactual value could affect other factors that could affect our outcome, or may cause our unit to leave the population. That may be want you want, but

- note that it is not about "holding all else constant"
- it sometimes engages absurdities

Example: what is the effect of leader gender on some outcome?

- For US presidents, this involves $Y_{trump}(female)$, i.e. "What would Donald Trump have done if he was a woman?'
- That is hard to imagine, depends on when you change his sex and how, affects many other characteristics, affects whether he would have been in sample of presidents, ...
- Raises questions as to what causal effect is of real interest.
- One guide for avoiding this is to assert that a cause must be "manipulable"
- IMHO: Good guidance, but non-manipulable things can have effects too.

Thinking about POs

- You are used to thinking about the observed Y_i. For example, you would not expect a causal effect if you found $Y \perp\!\!\!\perp D$.

Thinking about POs

- You are used to thinking about the observed Y_i. For example, you would not expect a causal effect if you found $Y \perp\!\!\!\perp D$.
- Remember, $Y_i(d)$ is not Y_i, it is "Y_i's value if-treated-with-d".

Thinking about POs

- You are used to thinking about the observed Y_i. For example, you would not expect a causal effect if you found $Y \perp\!\!\!\perp D$.
- Remember, $Y_i(d)$ is not Y_i, it is "Y_i's value if-treated-with-d".
- For example if you randomized D, you would see $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i$.

Thinking about POs

- You are used to thinking about the observed Y_i. For example, you would not expect a causal effect if you found $Y \perp\!\!\!\perp D$.
- Remember, $Y_i(d)$ is not Y_i, it is "Y_i's value if-treated-with-d".
- For example if you randomized D, you would see $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i$.
- Is this confusing? How can you digest this? Some tips:

Thinking about POs

- You are used to thinking about the observed Y_i. For example, you would not expect a causal effect if you found $Y \perp\!\!\!\perp D$.
- Remember, $Y_i(d)$ is not Y_i, it is "Y_i's value if-treated-with-d".
- For example if you randomized D, you would see $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i$.
- Is this confusing? How can you digest this? Some tips:
  - Read out $Y(1)$ as "the outcome you would get if treated", etc.

Thinking about POs

- You are used to thinking about the observed Y_i. For example, you would not expect a causal effect if you found $Y \perp\!\!\!\perp D$.
- Remember, $Y_i(d)$ is not Y_i, it is "Y_i's value if-treated-with-d".
- For example if you randomized D, you would see $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i$.
- Is this confusing? How can you digest this? Some tips:
  - Read out $Y(1)$ as "the outcome you would get if treated", etc.
  - D_i just switches whether you see $Y_i(1)$ or $Y_i(0)$, not affecting either

Thinking about POs

- You are used to thinking about the observed Y_i. For example, you would not expect a causal effect if you found $Y \perp\!\!\!\perp D$.
- Remember, $Y_i(d)$ is not Y_i, it is "Y_i's value if-treated-with-d".
- For example if you randomized D, you would see $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i$.
- Is this confusing? How can you digest this? Some tips:
  - Read out $Y(1)$ as "the outcome you would get if treated", etc.
  - D_i just switches whether you see $Y_i(1)$ or $Y_i(0)$, not affecting either
  - Note: Y_d is not *affected* by D. Thus, if Y_d and D were correlated, what would that mean about how treatment was assigned?

Brain Training

Test 1: Is $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_i|D_i = 1]$?

Brain Training

Test 1: Is $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_i|D_i = 1]$?

What licenses this?

Brain Training

Test 2: Suppose D_i randomly assigned.

Brain Training

Test 2: Suppose D_i randomly assigned.

- How would the Y_{1i} you observed (i.e. when $D_i = 1$) compare to the Y_{1i} you don't observe? And for Y_{0i}?

Brain Training

Test 2: Suppose D_i randomly assigned.

- How would the Y_{1i} you observed (i.e. when $D_i = 1$) compare to the Y_{1i} you don't observe? And for Y_{0i}?
- What can we say about $\mathbb{E}[Y_{1i}|D_i = 1]$ compared to $\mathbb{E}[Y_{1i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}]$?

Brain Training

Test 2: Suppose D_i randomly assigned.

- How would the Y_{1i} you observed (i.e. when $D_i = 1$) compare to the Y_{1i} you don't observe? And for Y_{0i}?
- What can we say about $\mathbb{E}[Y_{1i}|D_i = 1]$ compared to $\mathbb{E}[Y_{1i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}]$?

  (While we're here: this is ignorability, aka exchangeability!)

Brain Training

Test 2: Suppose D_i randomly assigned.

- How would the Y_{1i} you observed (i.e. when $D_i = 1$) compare to the Y_{1i} you don't observe? And for Y_{0i}?
- What can we say about $\mathbb{E}[Y_{1i}|D_i = 1]$ compared to $\mathbb{E}[Y_{1i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}]$?

  (While we're here: this is ignorability, aka exchangeability!)
- So can we estimate $\mathbb{E}[Y_{1i}]$? And $\mathbb{E}[Y_{0i}]$? How? What licenses do you use?

Brain Training

Test 3: Suppose $Y_{1i} = Y_{0i}$ for all i. But now, someone magically sees Y_{1i}, and chooses $D_i = 1$ more often when Y_{1i} is higher.

Brain Training

Test 3: Suppose $Y_{1i} = Y_{0i}$ for all i. But now, someone magically sees Y_{1i}, and chooses $D_i = 1$ more often when Y_{1i} is higher.

- How would the Y_{1i} you see compare to Y_{1i} you don't see?

Brain Training

Test 3: Suppose $Y_{1i} = Y_{0i}$ for all i. But now, someone magically sees Y_{1i}, and chooses $D_i = 1$ more often when Y_{1i} is higher.

- How would the Y_{1i} you see compare to Y_{1i} you don't see?
- So does $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_{1i}|D_i = 0] = \mathbb{E}[Y_{1i}]$?

Brain Training

Test 3: Suppose $Y_{1i} = Y_{0i}$ for all i. But now, someone magically sees Y_{1i}, and chooses $D_i = 1$ more often when Y_{1i} is higher.

- How would the Y_{1i} you see compare to Y_{1i} you don't see?
- So does $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_{1i}|D_i = 0] = \mathbb{E}[Y_{1i}]$?
- How would the Y_{0i} you see compare to the ones you don't?

Brain Training

Test 3: Suppose $Y_{1i} = Y_{0i}$ for all i. But now, someone magically sees Y_{1i}, and chooses $D_i = 1$ more often when Y_{1i} is higher.

- How would the Y_{1i} you see compare to Y_{1i} you don't see?
- So does $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_{1i}|D_i = 0] = \mathbb{E}[Y_{1i}]$?
- How would the Y_{0i} you see compare to the ones you don't?
- If you estimate $\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$, do you expect something near zero, positive, or negative?

Brain Training

Test 3: Suppose $Y_{1i} = Y_{0i}$ for all i. But now, someone magically sees Y_{1i}, and chooses $D_i = 1$ more often when Y_{1i} is higher.

- How would the Y_{1i} you see compare to Y_{1i} you don't see?
- So does $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_{1i}|D_i = 0] = \mathbb{E}[Y_{1i}]$?
- How would the Y_{0i} you see compare to the ones you don't?
- If you estimate $\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$, do you expect something near zero, positive, or negative?
- Does this tell us anything about $\mathbb{E}[Y_{1i} - Y_{0i}]$?

Brain Training

Test 3: Suppose $Y_{1i} = Y_{0i}$ for all i. But now, someone magically sees Y_{1i}, and chooses $D_i = 1$ more often when Y_{1i} is higher.

- How would the Y_{1i} you see compare to Y_{1i} you don't see?
- So does $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_{1i}|D_i = 0] = \mathbb{E}[Y_{1i}]$?
- How would the Y_{0i} you see compare to the ones you don't?
- If you estimate $\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$, do you expect something near zero, positive, or negative?
- Does this tell us anything about $\mathbb{E}[Y_{1i} - Y_{0i}]$?
- Can you say anything about expected direction of bias?

Brain Training

Test 4: Suppose you don't know anything about how D_i is assigned w.r.t. Y_{1i} and Y_{0i}.

Brain Training

Test 4: Suppose you don't know anything about how D_i is assigned w.r.t. Y_{1i} and Y_{0i}.

- Can we say anything about how $\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$ compares to $\mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}]$?

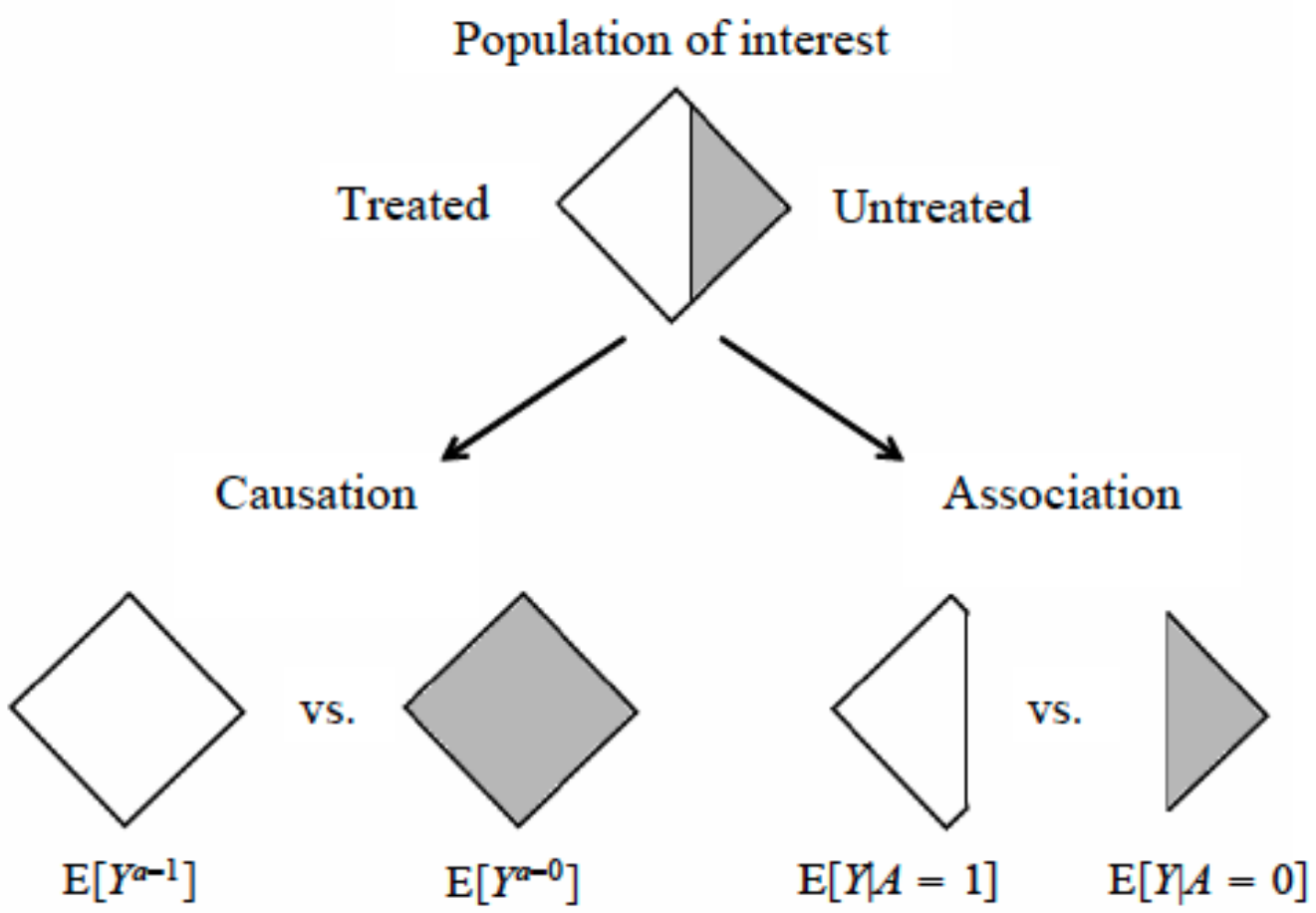
Brain Training

Test 4: Suppose you don't know anything about how D_i is assigned w.r.t. Y_{1i} and Y_{0i}.

- Can we say anything about how $\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$ compares to $\mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}]$?
- Unfortunately, this is often the scenario we are in with observational data.

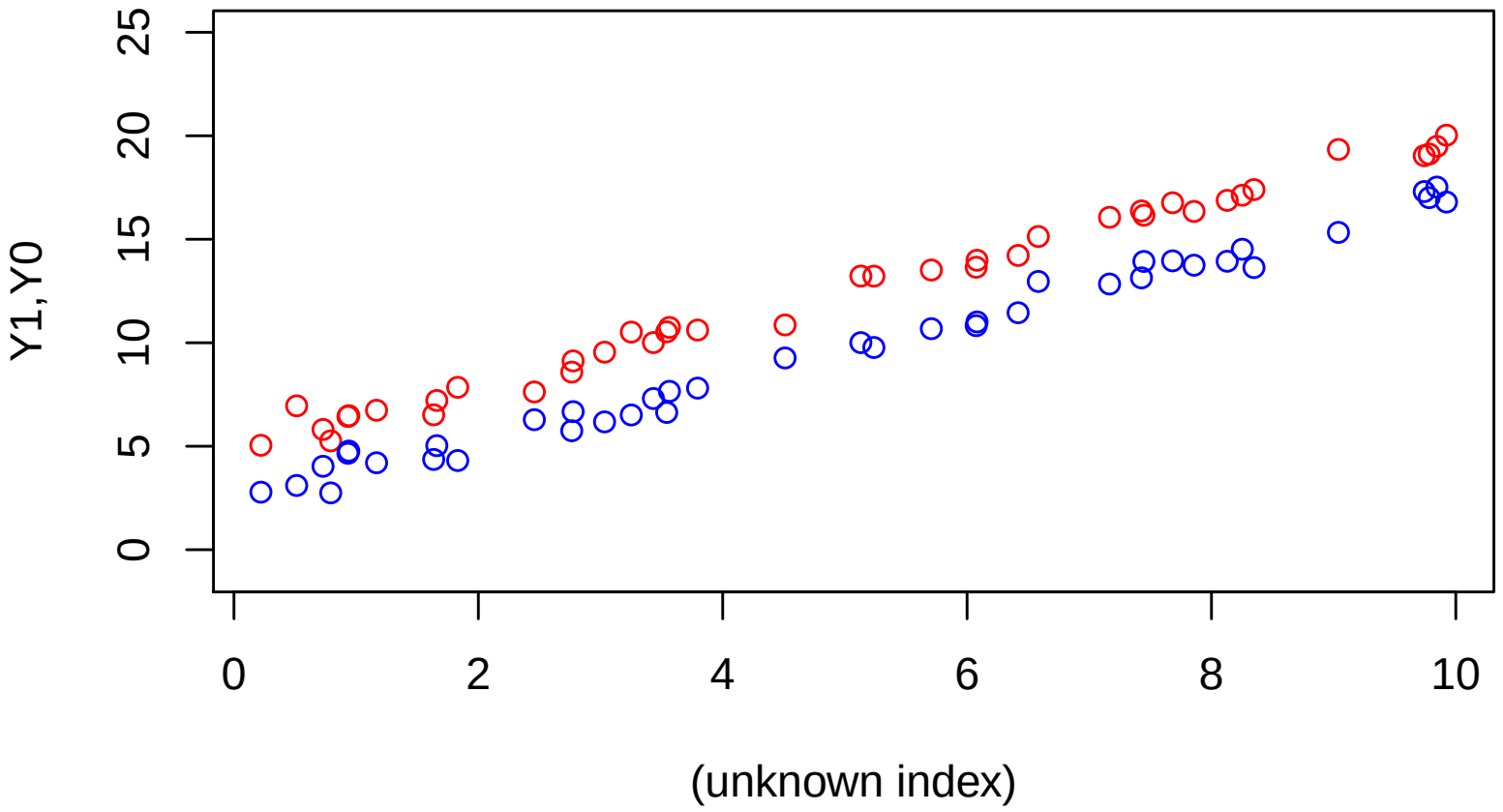
Brain Training: Visual Practice

Hernan & Robins figure 1.1:



Brain Training: Visual Practice

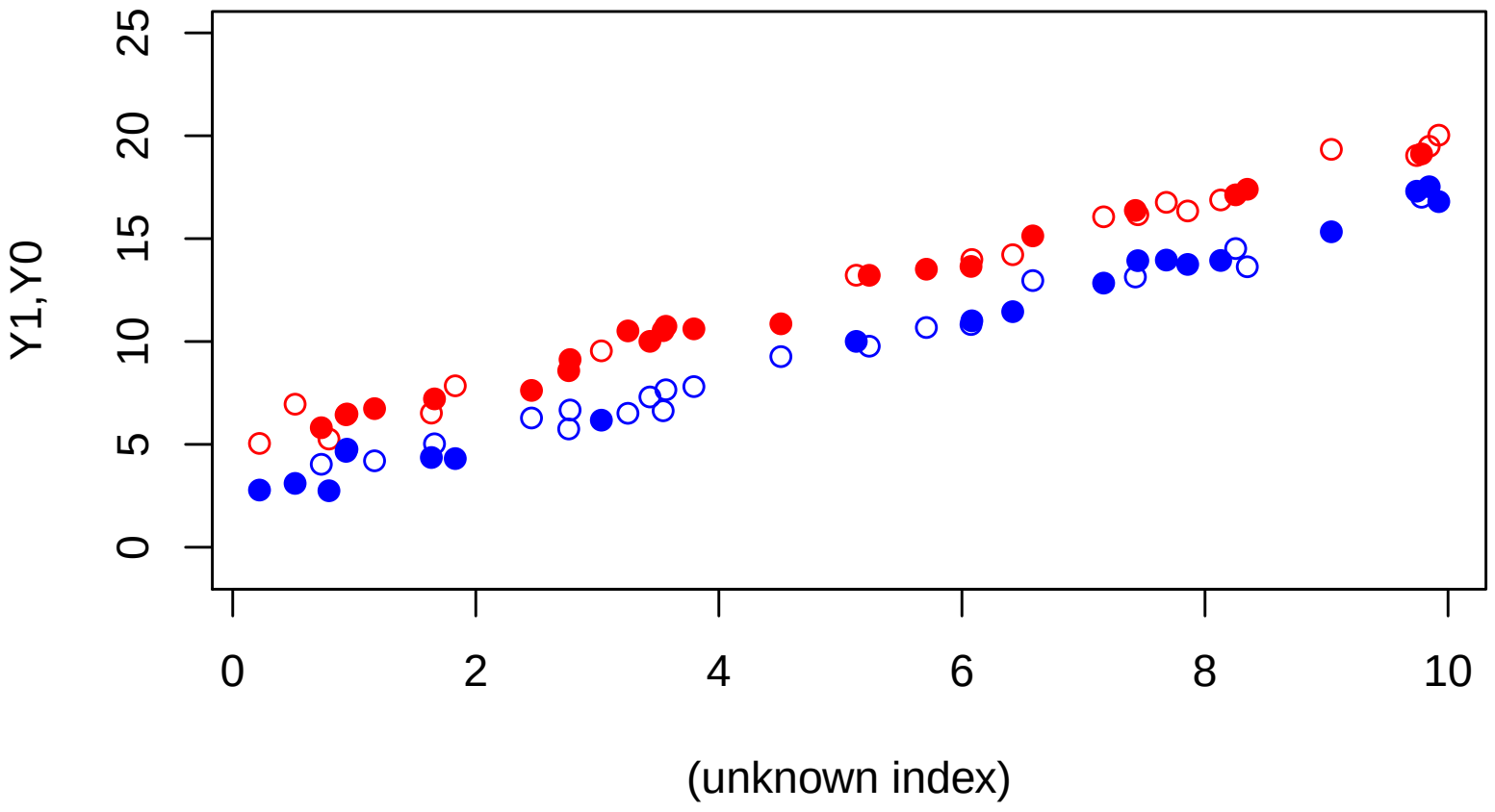
Let's looks at Y_{1i} and Y_{0i} alone first.



True $\mathbb{E}[Y_{1i} - Y_{0i}] = ATE = 3$

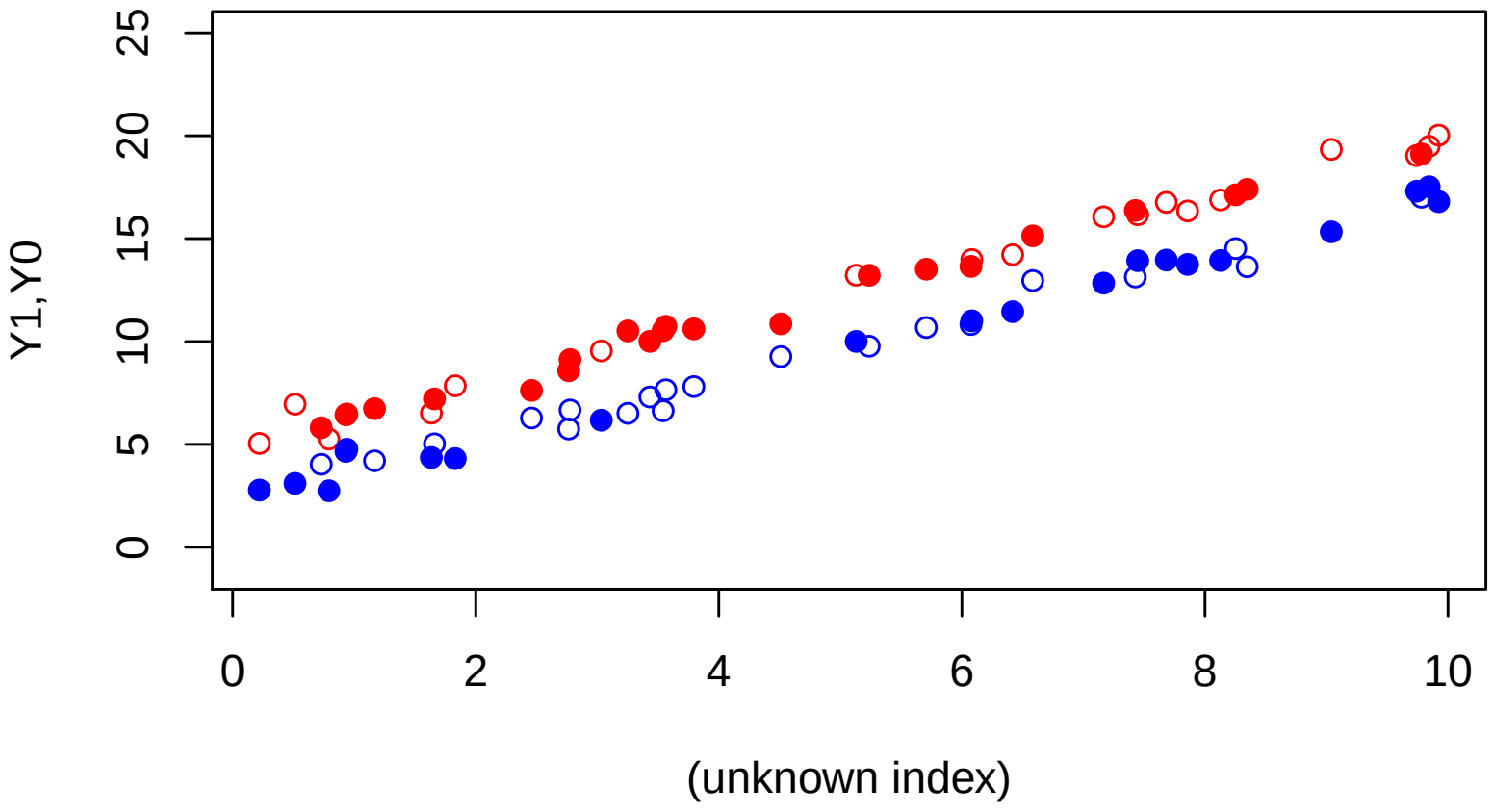
Random Assignment of Treatment

Now suppose $D_i = 1$ is randomly assigned. We see only the filled dots:



Random Assignment of Treatment

Now suppose $D_i = 1$ is randomly assigned. We see only the filled dots:

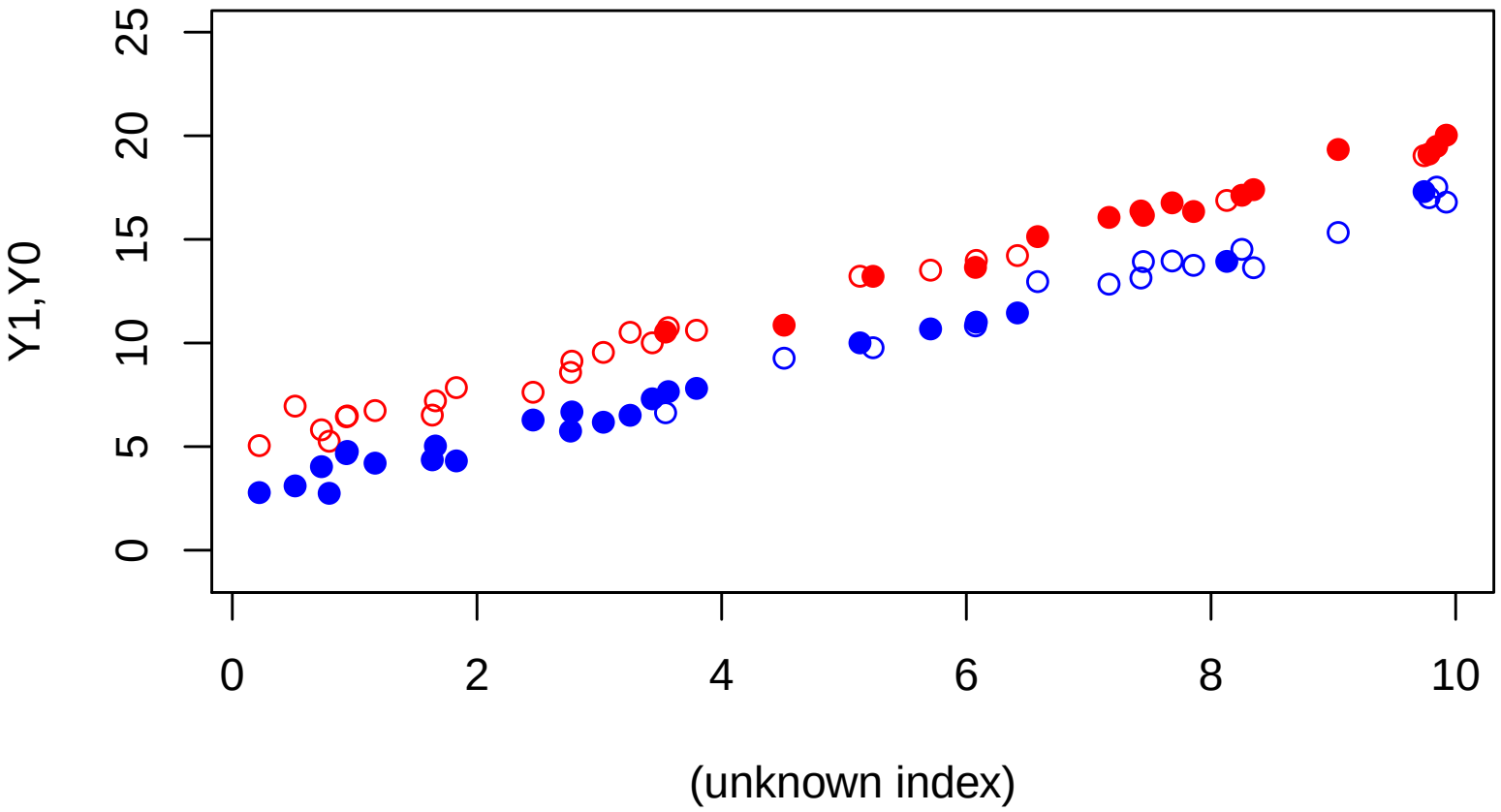


We get

$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = 3.01$$

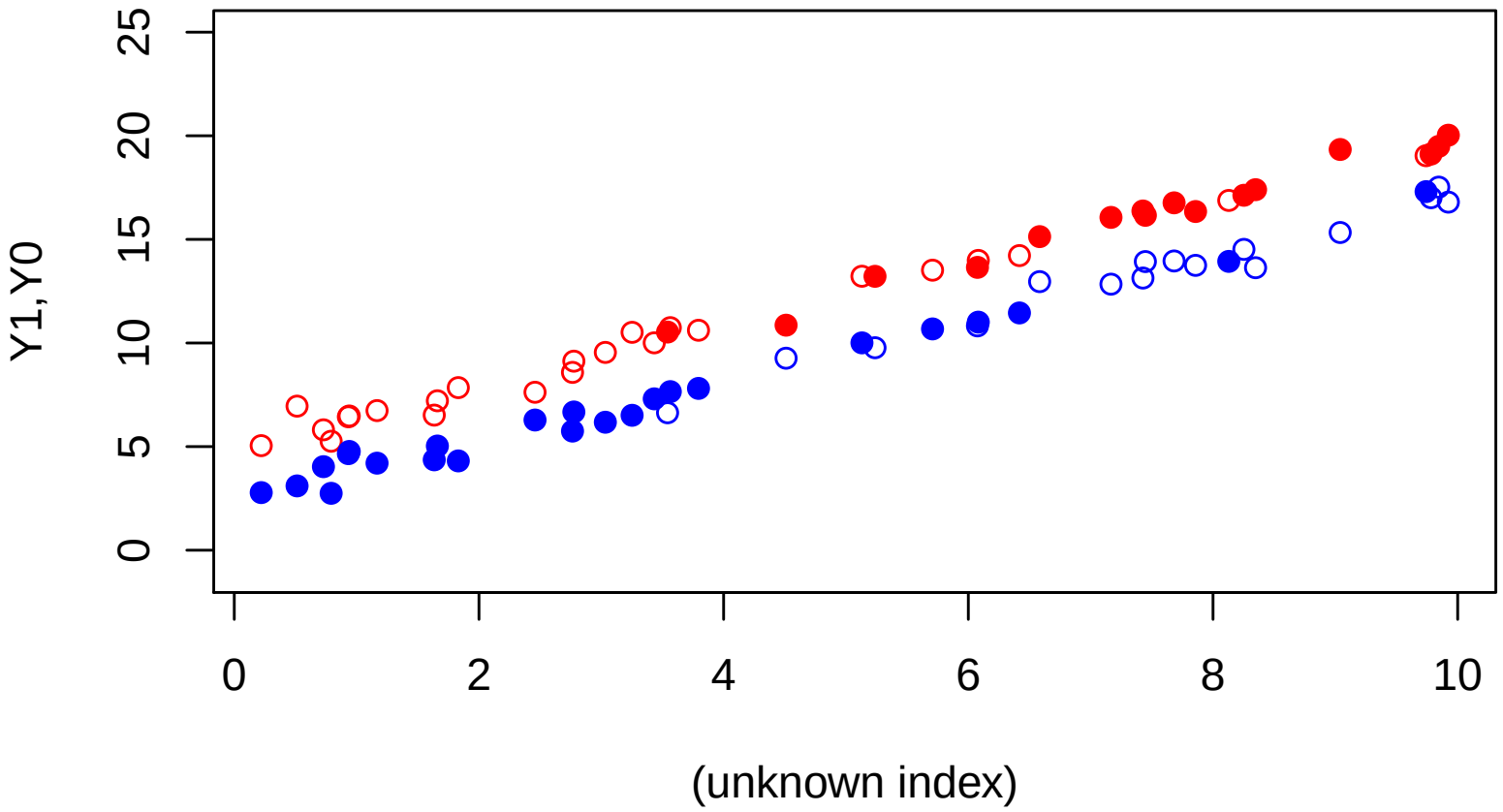
Non-Random Assignment of Treatment 1

Suppose $Pr(D_i = 1)$ increases to the right. Now we observe (filled dots):



Non-Random Assignment of Treatment 1

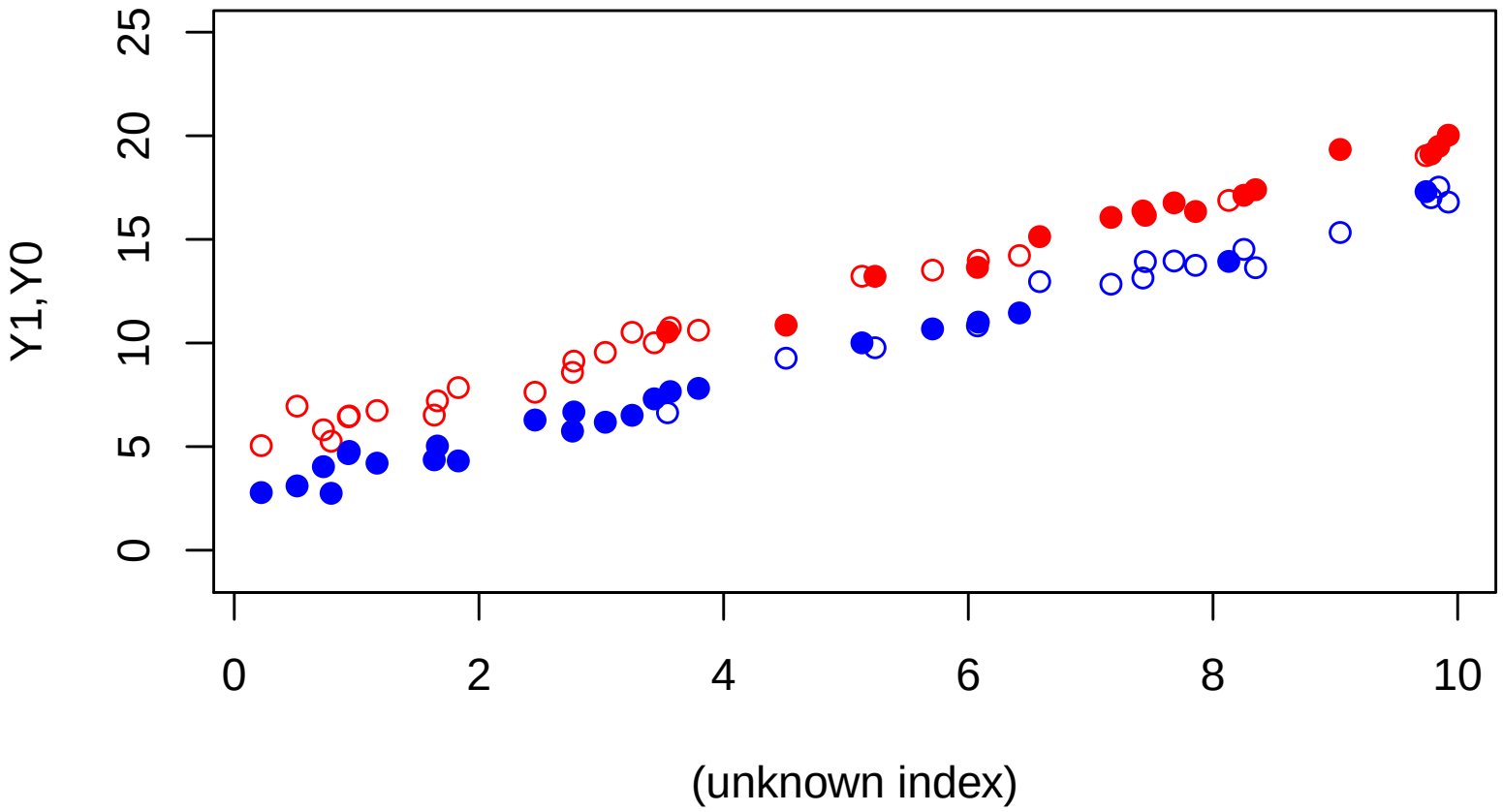
Suppose $Pr(D_i = 1)$ increases to the right. Now we observe (filled dots):



What difference in means do you expect?

Non-Random Assignment of Treatment 1

Suppose $Pr(D_i = 1)$ increases to the right. Now we observe (filled dots):

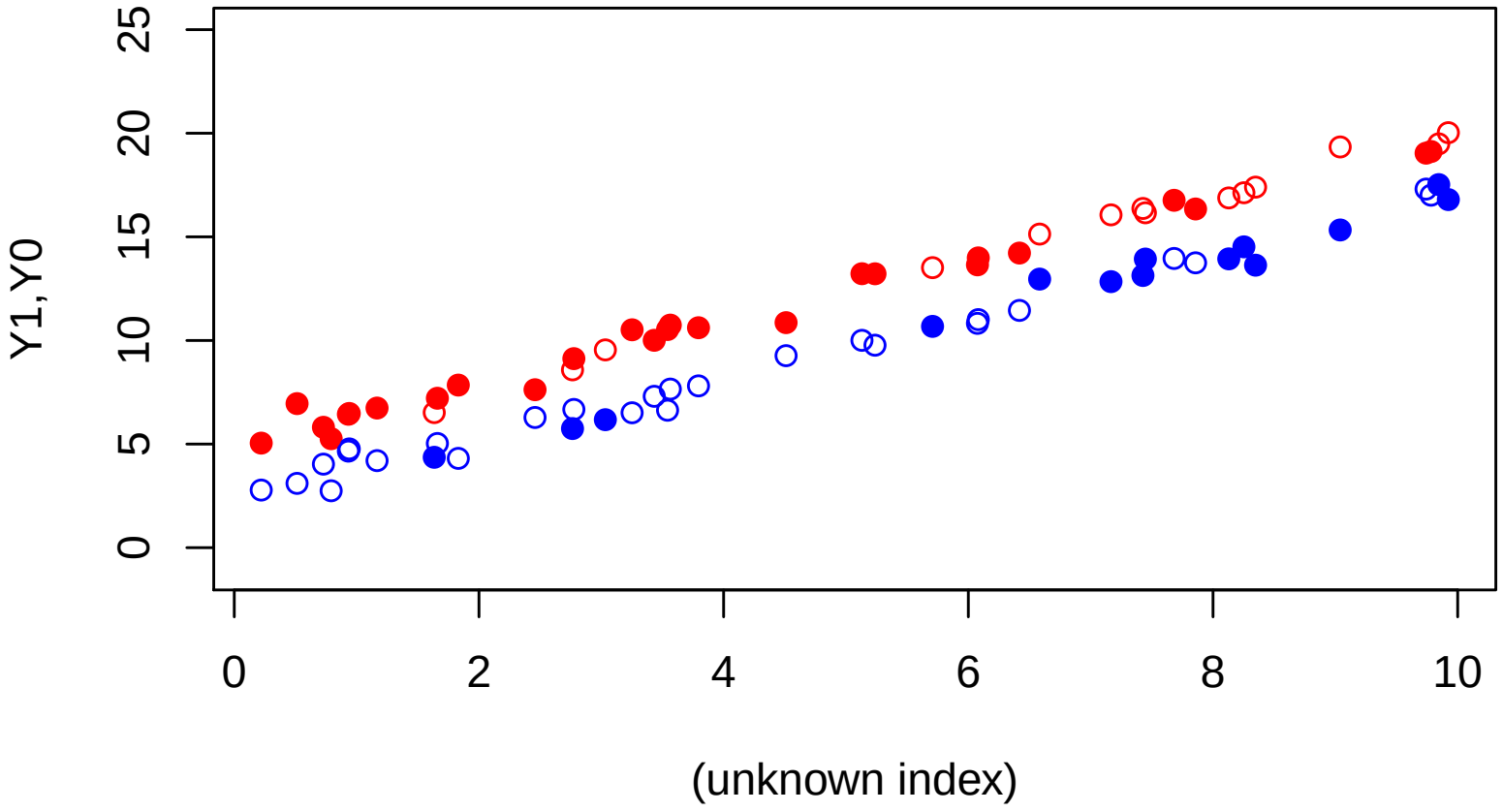


What difference in means do you expect?

$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = 9.97$$

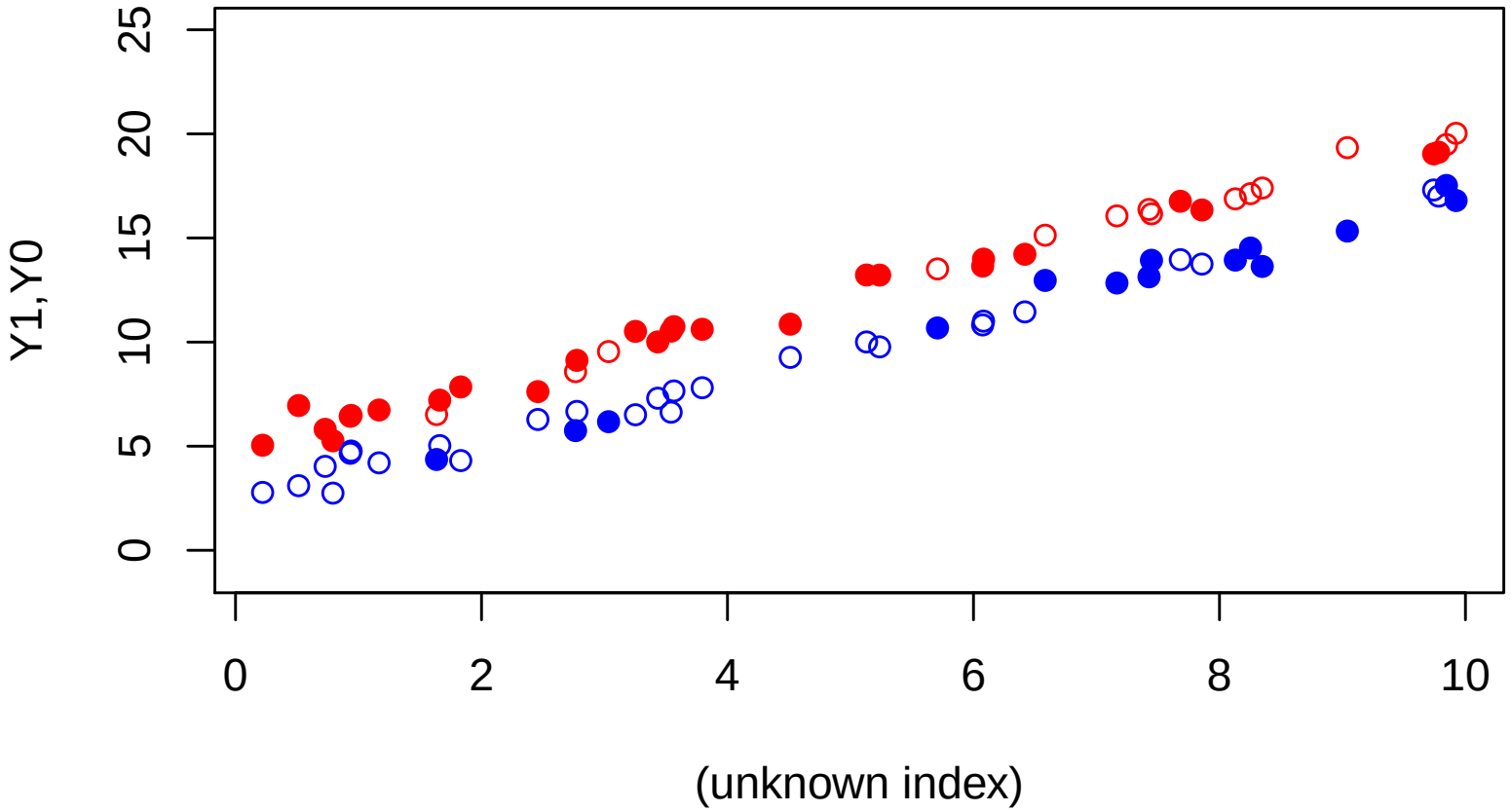
Non-Random Assignment of Treatment 2

Now suppose $Pr(D_i = 1)$ decreases to the right. We observe (filled dots):



Non-Random Assignment of Treatment 2

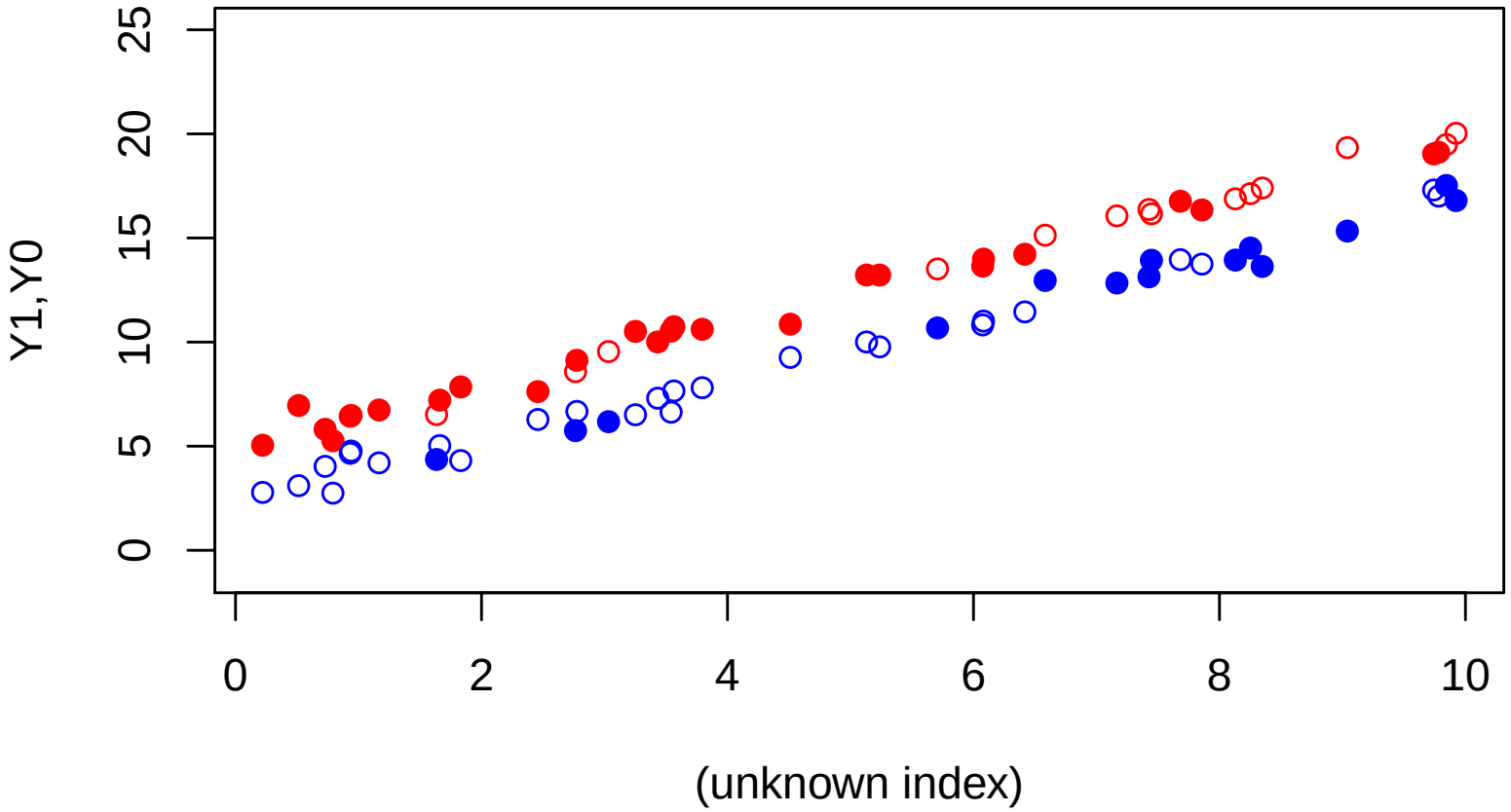
Now suppose $Pr(D_i = 1)$ decreases to the right. We observe (filled dots):



What difference in means do you expect now?

Non-Random Assignment of Treatment 2

Now suppose $Pr(D_i = 1)$ decreases to the right. We observe (filled dots):



What difference in means do you expect now?

$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = -1.59$$

Yet More Brain Training: “science” tables

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

Yet More Brain Training: "science" tables

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

1. What is the ATE?

Yet More Brain Training: "science" tables

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

1. What is the ATE? $\mathbb{E}[Y_{1i} - Y_{0i}] = 1/4 \times (6 - 1 + 0 + 3) = 2$

(Note: average effect is positive, but not all τ_i are)

Yet More Brain Training: "science" tables

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

1. What is the ATE? $\mathbb{E}[Y_{1i} - Y_{0i}] = 1/4 \times (6 - 1 + 0 + 3) = 2$

(Note: average effect is positive, but not all τ_i are)

2. What are the ATT and ATC?

Yet More Brain Training: "science" tables

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

1. What is the ATE? $\mathbb{E}[Y_{1i} - Y_{0i}] = 1/4 \times (6 - 1 + 0 + 3) = 2$

(Note: average effect is positive, but not all τ_i are)

2. What are the ATT and ATC?

$$\mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1] = .5(6 - 1) = 2.5$$

Yet More Brain Training: "science" tables

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

1. What is the ATE? $\mathbb{E}[Y_{1i} - Y_{0i}] = 1/4 \times (6 - 1 + 0 + 3) = 2$

(Note: average effect is positive, but not all τ_i are)

2. What are the ATT and ATC?

$$\mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1] = .5(6 - 1) = 2.5$$

$$\mathbb{E}[Y_{1i} - Y_{0i} | D_i = 0] = .5(0 + 3) = 1.5$$

Difference in Means

You saw earlier how simple comparison of observed outcomes can be misleading. Let's use POM to see this more rigorously.

Difference in Means

You saw earlier how simple comparison of observed outcomes can be misleading. Let's use POM to see this more rigorously.

The Difference in Means estimand:

$$\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$$

Difference in Means

You saw earlier how simple comparison of observed outcomes can be misleading. Let's use POM to see this more rigorously.

The Difference in Means estimand:

$$\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$$

What does the POM tell us about this quantity? One decomposition:

$$\begin{aligned}\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] &= \mathbb{E}[Y_{1i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0] \\ &= \mathbb{E}[Y_{1i} - Y_{0i}|D_i = 1] + (\mathbb{E}[Y_{0i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0])\end{aligned}$$

Difference in Means

You saw earlier how simple comparison of observed outcomes can be misleading. Let's use POM to see this more rigorously.

The Difference in Means estimand:

$$\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$$

What does the POM tell us about this quantity? One decomposition:

$$\begin{aligned}\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] &= \mathbb{E}[Y_{1i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0] \\ &= \mathbb{E}[Y_{1i} - Y_{0i}|D_i = 1] + (\mathbb{E}[Y_{0i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0])\end{aligned}$$

What are the terms in blue & green?

The Assignment Mechanism

To fill in the missing potential outcomes using observed ones, assumptions about the *assignment mechanism* come into play.

The Assignment Mechanism

To fill in the missing potential outcomes using observed ones, assumptions about the *assignment mechanism* come into play.

For example, the bias we just saw implies we can get *ATT* when $Y_{0i} \perp D_i$

- (This is a special case of the broader assumption, $\{Y_0, Y_1\} \perp\!\!\!\perp D_i$, we will discuss with experiments)
- Point is: if we have reason to believe $D_i \perp\!\!\!\perp \{Y_0, Y_1\}$, we know we can substitute group means for everybody's mean on the corresponding potential outcome.
- This counts as an "identification strategy" or "identifying assumption" since it allows us to link an observable thing with a counterfactual thing.

Summing Up: The Neyman-Rubin / Potential Outcome Model

- Defines causality through counterfactual comparisons

Summing Up: The Neyman-Rubin / Potential Outcome Model

- Defines causality through counterfactual comparisons
- In so doing, forces us to think about potential outcomes rather than just the observed outcome.

Summing Up: The Neyman-Rubin / Potential Outcome Model

- Defines causality through counterfactual comparisons
- In so doing, forces us to think about potential outcomes rather than just the observed outcome.
- Consistency, no spillovers/interference, one-version-of-treatment implied by notation.

Summing Up: The Neyman-Rubin / Potential Outcome Model

- Defines causality through counterfactual comparisons
- In so doing, forces us to think about potential outcomes rather than just the observed outcome.
- Consistency, no spillovers/interference, one-version-of-treatment implied by notation.
- Some dangers:
  - violating consistency/ spillover, multiple versions of treatment
  - POs you wouldn't actually care about when thinking of a treatment.
- There is no progress without estimating missing potential outcomes.

Summing Up: The Neyman-Rubin / Potential Outcome Model

- Defines causality through counterfactual comparisons
- In so doing, forces us to think about potential outcomes rather than just the observed outcome.
- Consistency, no spillovers/interference, one-version-of-treatment implied by notation.
- Some dangers:
  - violating consistency/ spillover, multiple versions of treatment
  - POs you wouldn't actually care about when thinking of a treatment.
- There is no progress without estimating missing potential outcomes.
- Whether and how we estimate missing potential outcomes will have to do with what assumptions we can make about the assignment mechanism.